INCH POUND
MIL-M-38510/765
AMENDMENT 1
26 March 1998

MILITARY SPECIFICATION

MICROCIRCUITS, DIGITAL, ADVANCED CMOS, SHIFT REGISTERS, MONOLITHIC SILICON, POSITIVE LOGIC

Inactive for new design after 9 August 1996

This amendment forms a part of MIL-M-38510/765, dated 25 February 1991, and is approved for use by all Departments and Agencies of the Department of Defense.

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Table I, high level output voltage, V_{OH3}, device type column, RHA levels M, D, and R; add “06”.
Table I, high level output voltage, V_{OH3}, -55°C/+25°C limits, min column, RHA levels M, D, and R; add “5.4”.

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Table I, high level output voltage, V_{OH5}, device type column, RHA levels M, D, and R; add “06”.
Table I, high level output voltage, V_{OH5}, -55°C/+25°C limits min column, RHA levels M, D, and R; add “3.7”.
Table I, high level output voltage, V_{OH7}, device type column, RHA levels M, D, and R; add “06”.
Table I, high level output voltage, V_{OH7}, -55°C/+25°C limits min column, RHA levels M, D, and R; add “3.85”.

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Table I, low level output voltage, V_{OL3}, device type column, RHA levels M, D, and R; add “06”.
Table I, low level output voltage, V_{OL3}, -55°C/+25°C limits max column, RHA levels M, D, and R; add “0.1”.
Table I, low level output voltage, V_{OL5}, device type column, RHA levels M, D, and R; add “06”.
Table I, low level output voltage, V_{OL5}, -55°C/+25°C limits max column, RHA levels M, D, and R; add “0.4”.

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Table I, low level output voltage, V_{OL7}, device type column, RHA levels M, D, and R; add “06”.
Table I, low level output voltage, V_{OL7}, -55°C/+25°C limits max column, RHA levels M, D, and R; add “1.65”.
Table I, positive input clamp voltage, V_{IC+}, device type column, RHA levels M, D, and R; add “06”.
Table I, positive input clamp voltage, V_{IC+}, -55°C/+25°C limits min column, RHA levels M, D, and R; add “0.4”.
Table I, positive input clamp voltage, V_{IC+}, -55°C/+25°C limits max column, RHA levels M, D, and R; add “1.5”.

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Table I, negative input clamp voltage, V_{IC-}, device type column, RHA levels M, D, and R; add “06”.
Table I, negative input clamp voltage, V_{IC-}, -55°C/+25°C limits min column, RHA levels M, D, and R; add “-0.4”.
Table I, negative input clamp voltage, V_{IC-}, -55°C/+25°C limits max column, RHA levels M, D, and R; add “-1.5”.
Table I, input current low, I_{IL}, device type column, RHA levels M, D, and R; add “06”.
Table I, input current low, I_{IL}, -55°C/+25°C limits max column, RHA levels M, D, and R; add “-0.1”.
Table I, input current high, I_{IH}, device type column, RHA levels M, D, and R; add “06”.
Table I, input current high, I_{IH}, -55°C/+25°C limits max column, RHA levels M, D, and R; add “0.1”.

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Table I, supply current quiescent, I_{CC}, device type column, RHA levels M, D, and R; add “06”.
Table I, supply current quiescent, I_{CC}, -55°C/+25°C limits max column, RHA level M; add “25”.
Table I, supply current quiescent, I_{CC}, -55°C/+25°C limits max column, RHA level D; add “200”.
Table I, supply current quiescent, I_{CC}, -55°C/+25°C limits max column, RHA level R; add “700”.
Table I, three-state output leakage current low, I_{OZL}, device type column, RHA levels M, D, and R; add “06”.
Table I, three-state output leakage current low, I_{OZL}, -55°C/+25°C limits max column, RHA levels M, D, and R; add “-0.6”.
Table I, three-state output leakage current high, I_{OZH}, device type column, RHA levels M, D, and R; add “06”.
Table I, three-state output leakage current high, I_{OZH}, -55°C/+25°C limits max column, RHA levels M, D, and R; add “+0.6”.

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Table I, power dissipation capacitance, C_{PD}, -55°C/+25°C limits max column; delete “190” and replace with “250”.

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Table I, truth table test, delete and replace with the following:

Test	Symbol	Test conditions, unless otherwise specified 1/		Device type 2/	V_{CC}	Group A subgroups (test method)	-55°C/+25°C limits		+125°C limits		Unit
							Min	Max	Min 3/	Max 3/	
Truth table test	12/	V_{IL} = .45 V V_{IH} = 2.5 V Verify output V_O	M	06	3.0 V	7 (3014) 5/	L	H			
			D				L	H			
			R				L	H			
		V_{IL} = .6 V V_{IH} = 3.7 V Verify output V_O		All	4.5 V	7, 8 (3014)	L	H	L	H	

Table I, propagation delay time, CLK to I/On, t_{PHL1}, t_{PLH1}, V_{CC} = 3.0 V, device type column, RHA levels M, D, and R; add “06”.
Table I, propagation delay time, CLK to I/On, t_{PHL1}, t_{PLH1}, V_{CC} = 3.0 V, -55°C/+25°C limits min column, RHA levels M, D, and R; add “1.0”.
Table I, propagation delay time, CLK to I/On, t_{PHL1}, t_{PLH1}, V_{CC} = 3.0 V, -55°C/+25°C limits max column, RHA levels M, D, and R; add “23.0”.
Table I, propagation delay time, CLK to I/On, t_{PHL1}, t_{PLH1}, V_{CC} = 4.5 V, device type column, RHA levels M, D, and R; add “06”.
Table I, propagation delay time, CLK to I/On, t_{PHL1}, t_{PLH1}, V_{CC} = 4.5 V, -55°C/+25°C limits min column, RHA levels M, D, and R; add “1.0”.
Table I, propagation delay time, CLK to I/On, t_{PHL1}, t_{PLH1}, V_{CC} = 4.5 V, -55°C/+25°C limits max column, RHA levels M, D, and R; add “16.0”.

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Table I, propagation delay time, CLK to Q0, CLK to Q7, t_{PHL2}, t_{PLH2}, V_{CC} = 3.0 V, device type column, RHA levels M, D, and R; add “06”.
Table I, propagation delay time, CLK to Q0, CLK to Q7, t_{PHL2}, t_{PLH2}, V_{CC} = 3.0 V, -55°C/+25°C limits min column, RHA levels M, D, and R; add “1.0”.
Table I, propagation delay time, CLK to Q0, CLK to Q7, t_{PHL2}, t_{PLH2}, V_{CC} = 3.0 V, -55°C/+25°C limits max column, RHA levels M, D, and R; add “21.5”.
Table I, propagation delay time, CLK to Q0, CLK to Q7, t_{PHL2}, t_{PLH2}, V_{CC} = 4.5 V, device type column, RHA levels M, D, and R; add “06”.
Table I, propagation delay time, CLK to Q0, CLK to Q7, t_{PHL2}, t_{PLH2}, V_{CC} = 4.5 V, -55°C/+25°C limits min column, RHA levels M, D, and R; add “1.0”.
Table I, propagation delay time, CLK to Q0, CLK to Q7, t_{PHL2}, t_{PLH2}, V_{CC} = 4.5 V, -55°C/+25°C limits max column, RHA levels M, D, and R; add “14.5”.

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Table I, propagation delay time, $\overline{MR}$ to Q0, $\overline{MR}$ to Q7, t_{PHL3}, V_{CC} = 3.0 V, device type column, RHA levels M, D, and R; add “06”.
Table I, propagation delay time, $\overline{MR}$ to Q0, $\overline{MR}$ to Q7, t_{PHL3}, V_{CC} = 3.0 V, -55°C/+25°C limits min column, RHA levels M, D, and R; add “1.0”.
Table I, propagation delay time, $\overline{MR}$ to Q0, $\overline{MR}$ to Q7, t_{PHL3}, V_{CC} = 3.0 V, -55°C/+25°C limits max column, RHA levels M, D, and R; add “22.5”.
Table I, propagation delay time, $\overline{MR}$ to Q0, $\overline{MR}$ to Q7, t_{PHL3}, V_{CC} = 4.5 V, device type column, RHA levels M, D, and R; add “06”.
Table I, propagation delay time, $\overline{MR}$ to Q0, $\overline{MR}$ to Q7, t_{PHL3}, V_{CC} = 4.5 V, -55°C/+25°C limits min column, RHA levels M, D, and R; add “1.0”.
Table I, propagation delay time, $\overline{MR}$ to Q0, $\overline{MR}$ to Q7, t_{PHL3}, V_{CC} = 4.5 V, -55°C/+25°C limits max column, RHA levels M, D, and R; add “15.5”.

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Table I, propagation delay time, $\overline{MR}$ to I/0n, t_{PHL4}, V_{CC} = 3.0 V, device type column, RHA levels M, D, and R; add “06”.
Table I, propagation delay time, $\overline{MR}$ to I/0n, t_{PHL4}, V_{CC} = 3.0 V, -55°C/+25°C limits min column, RHA levels M, D, and R; add “1.0”.
Table I, propagation delay time, $\overline{MR}$ to I/0n, t_{PHL4}, V_{CC} = 3.0 V, -55°C/+25°C limits max column, RHA levels M, D, and R; add “21.5”.
Table I, propagation delay time, $\overline{MR}$ to I/0n, t_{PHL4}, V_{CC} = 4.5 V, device type column, RHA levels M, D, and R; add “06”.
Table I, propagation delay time, $\overline{MR}$ to I/0n, t_{PHL4}, V_{CC} = 4.5 V, -55°C/+25°C limits min column, RHA levels M, D, and R; add “1.0”.
Table I, propagation delay time, $\overline{MR}$ to I/0n, t_{PHL4}, V_{CC} = 4.5 V, -55°C/+25°C limits max column, RHA levels M, D, and R; add “15.0”.

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Table I, output enable time, $\overline{OE}$n to I/On, t_{PZH1}, t_{PZL1}, V_{CC} = 3.0 V, device type column, RHA levels M, D, and R; add “06”.
Table I, output enable time, $\overline{OE}$n to I/On, t_{PZH1}, t_{PZL1}, V_{CC} = 3.0 V, -55°C/+25°C limits min column, RHA levels M, D, and R; add “1.0”.
Table I, output enable time, $\overline{OE}$n to I/On, t_{PZH1}, t_{PZL1}, V_{CC} = 3.0 V, -55°C/+25°C limits max column, RHA levels M, D, and R; add “19.5”.
Table I, output enable time, $\overline{OE}$n to I/On, t_{PZH1}, t_{PZL1}, V_{CC} = 4.5 V, device type column, RHA levels M, D, and R; add “06”.

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Table I, output enable time, $\overline{OE}n$ to I/On, t_{PZH1}, t_{PZL1}, V_{CC} = 4.5 V, -55°C/+25°C limits min column, RHA levels M, D, and R; add “1.0”.
Table I, output enable time, $\overline{OE}n$ to I/On, t_{PZH1}, t_{PZL1}, V_{CC} = 4.5 V, -55°C/+25°C limits max column, RHA levels M, D, and R; add “13.0”.

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Delete “15/ Measured only for initial test and after process or design changes which may affect this parameter.”

CONCLUDING MATERIAL

Custodians:
Army - CR
Navy - EC
Air Force - 17

NASA - NA

Preparing activity:
DLA - CC

Review activities:
Navy - OS, TD, AS, CG, MC, SH
Army - AR, MI, SM
Air Force - 19, 85, 99

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